

Video 13 – Validation

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Video Synopsis

Introduction **03:42 – 07:18**

Validation is contrasted with regularization and it is shown that where regularization tries to limit the augmented error, validation estimates E_{out} directly.

SAFE SKIP! **Mathematical prove of relationship E_{val} and E_{out}** **07:18-14:15**

Prof. Abu Mostafa shows that the error obtained in validation becomes a better estimate of E_{out} as the number of data points used for validation (K) increases.

The validation dilemma **14:15 -23:50**

In this section the inherent dilemma in validation is explained: you want a high K to get a good estimate of your out of sample error for the final hypothesis, but by taking K samples away from your dataset, you have a fewer data points (N) to train on, which results in your final hypothesis being suboptimal! In other words, with high K you very accurately predict the out-of-sample error for a hypothesis with high in-sample error. With low K you get a bad estimate of the out-of-sample error for a hypothesis with low in-sample error. Neither is satisfactory..

The essential difference between validation and testing **23:50 – 30:49**

The essential difference between validation and testing is explained: validation, because we use it to choose the most optimistic estimate of E_{out} , is biased, i.e. it is always optimistic about the out-of-sample error that we should expect. This is exemplified using an example with two hypotheses, each having an out-of-sample (expected) error of 0.5. When you now use valuation to pick the best of these two hypotheses, you will pick the one that happens to have the lowest E_{out} . The chance that you find an E_{out} lower than 0.5 is 75%. This shows that the E_{out} after valuation is biased optimistically. The exact same holds for early stopping in validation for neural nets. It is argued that this bias is always present in validation, but in general it will be low enough to be ignored.

Model selection **30:49 – 40:16**

Model selection is introduced as the choice for a certain hypothesis in the widest sense. Model selection can pertain to making a choice between linear classifiers or more complex classifiers, number of features, or just the choice of λ for the regularisation capability of a model. The section goes on to argue, that by choosing a certain model, the estimate of the out-of-sample error for that model is biased because you chose that model (and its error) because the error was

low! The section ends with an informative graph showing that E_{out} increases as K increases (because N , the number of training sample, decreases with increasing K) and that the estimate obtained of E_{out} through validation improves with increasing K .

Validation in light of the theory of generalization

40:16 – 49:06

This is a really interesting section in which validation is linked back to the theory of generalisation and the VC bound. In particular, this section shows that we can think of the error made in E_{val} (relative to E_{out}) as an error that will increase with an increasing number of models to choose from (M) and will decrease if we have more data to validate. This is exactly what we saw when talking about the VC bound and the same types of rules of thumb hold: the number of validation samples you have, determines the number of parameters you can tune using validation. The section ends with a short discussion on contamination in the sense that a data set can no longer be used to predict E_{out} accurately. Your training data set is heavily contaminated: it provides a really low E_{in} , but is really bad at predicting E_{out} . A test set is very good at predicting E_{out} as it has never been used on the model, i.e. it is completely uncontaminated. The validation set will fall in between: it will be slightly contaminated once you use it to tune one parameter. For every further parameter it will be contaminated more and the data set's ability to predict E_{out} accurately will suffer! The lesson is that you should be conservative with the number of parameters you tune with any one validation data set.

The validation dilemma solved – Leave one out

49:06 – 57:37

The dilemma for the choice of K is repeated (you want high K to make E_{val} a good estimate of E_{out} , but you also want K small so that N is big enough to choose the best hypothesis) and the solution to this seemingly paradoxical request is provided: Leave one out. It is shown that you can take the full data set minus one point for training and use only one point for validation. By repeating this process N times (using each data point out of the data set of N points once for validation) and averaging the found validation errors, you get a good estimate of E_{out} ! This error is called the cross-validation error, E_{cv} . As this error is made up of N different validation exercises, you can think of E_{cv} as originating from a validation with N points with the training performed on $N-1$ points. Hence we have achieved what we set out to do: the training set is $N-1$ samples and the validation set (K) N samples.

An example of cross validation

57:37 - 01:02:14

An example is given of cross validation on a data set of 3 points showing that cross validation will result in the choice of the most suitable hypothesis. This example also illustrates the process used in cross validation.

An example of validation showing the parallels with regularization

01:02:14 – 01:06:22

A more realistic example is given to demonstrate the use of validation for feature selection and to show that it achieves the exact same goal as regularisation, but now by estimating E_{out} through E_{val} and then making model choices accordingly.

K-fold cross validation

01:06:22 – 01:08:34

K -fold cross validation is introduced as an extension to leave-one-out validation in which not one point is used for validation, but a certain fraction of the data, for example $1/10^{\text{th}}$ of the data. This is desirable as Leave-one-out validation results in a model being trained N times, which becomes unfeasible for large N . In case of 10-fold cross validation, you divide your data set into 10 equal parts, you take 9 parts for training and one for validation and carry on exactly as done before. But instead of training the model N times (for Leave-one-out), you now only have to train the model 10 times.

SAFE SKIP! Q&A

01:08:34-end